ACP Summer School 2026

CP: Explanation, Verification and Applications

2026年吉林大学计算机科学拔尖人才培养基地“成长伙伴”国际暑假学校

Optimisation Challenge: Airport Turnaround Scheduling

Changchun, China
24 – 28 August 2026

Why Hackathon?

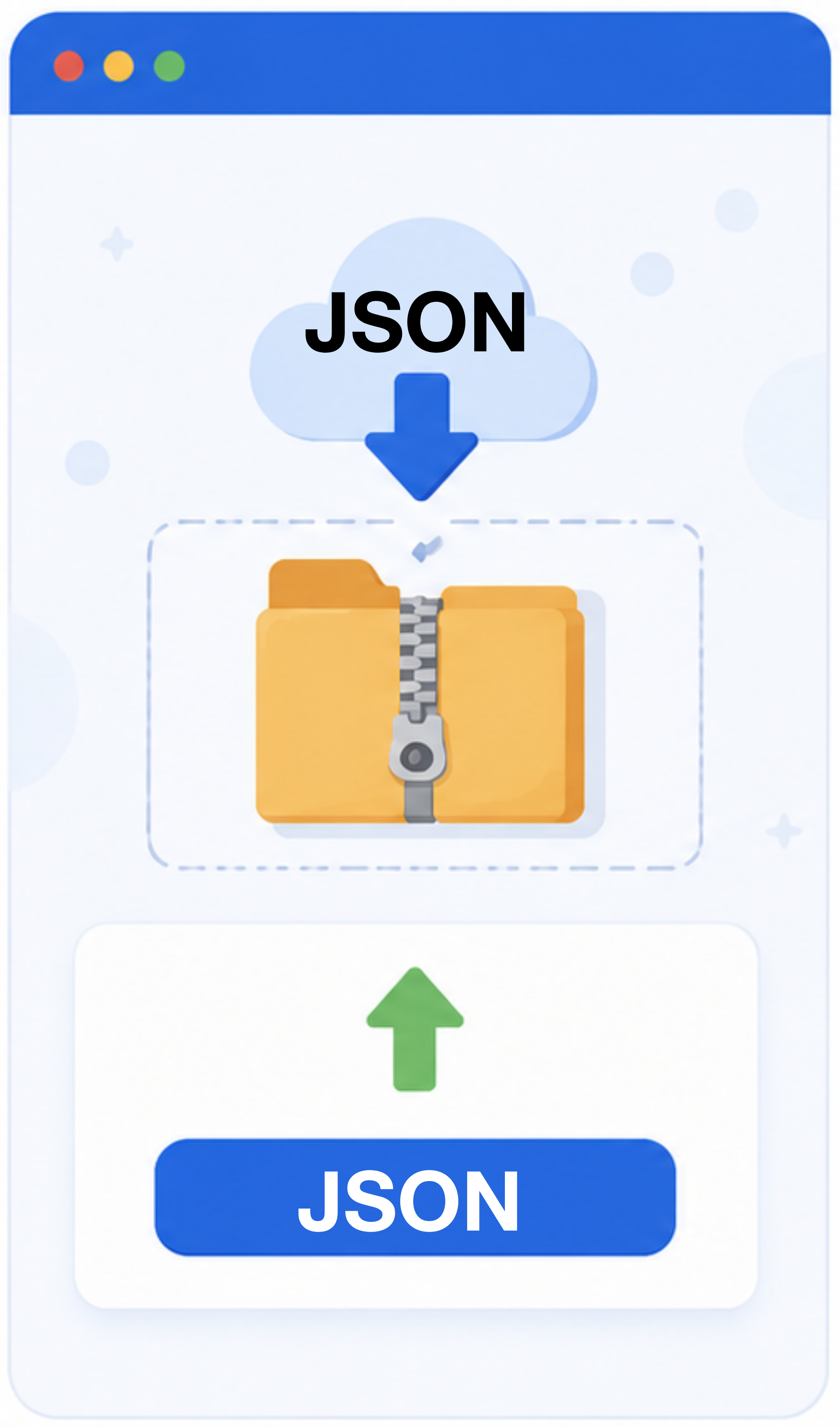
- Learn swimming in the swimming pool
- Tackle a real-ish optimisation problem
- Meet fellow optimisers and make friends
- Choose any solving technology you like

Turnaround Scheduling

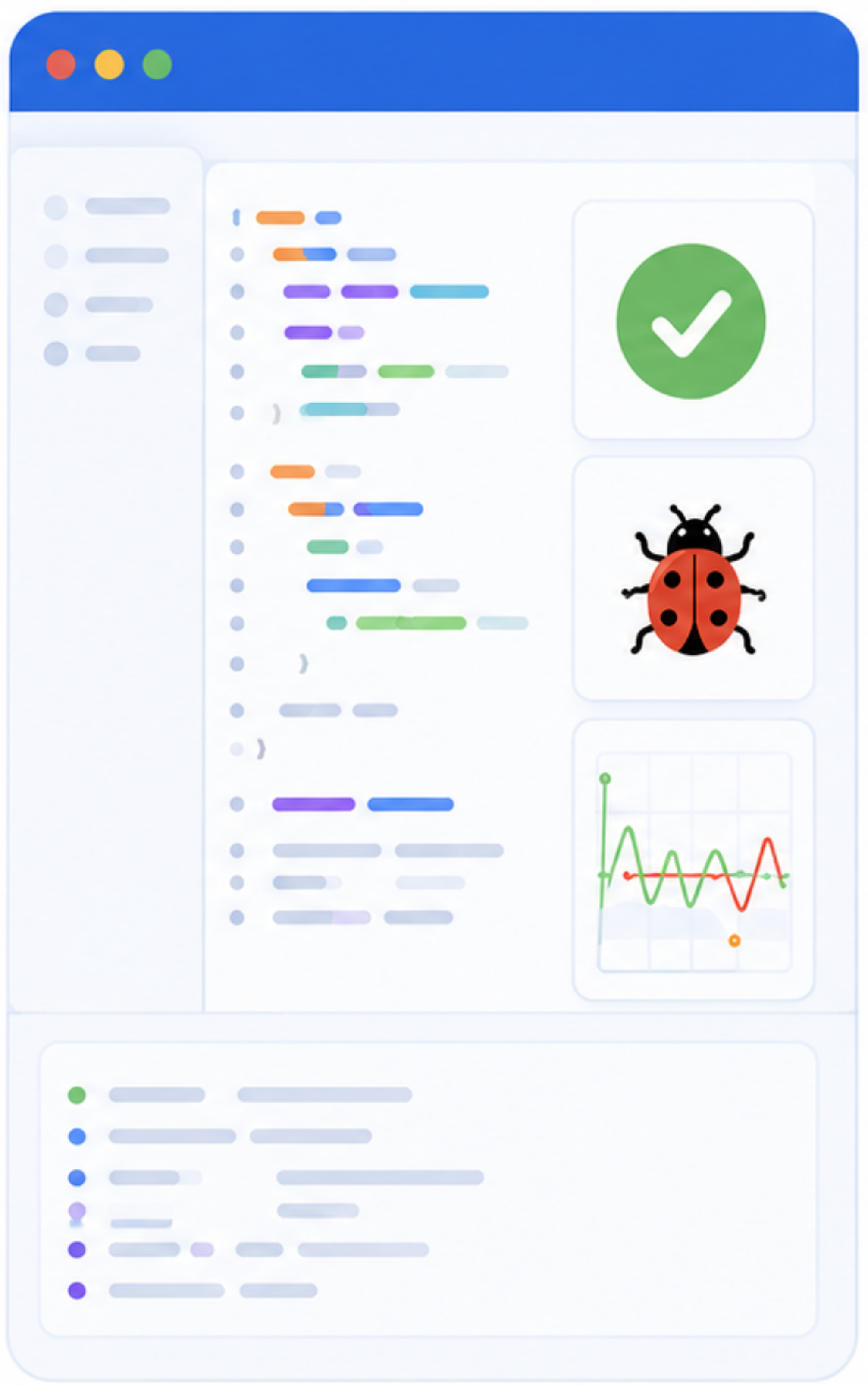
- Daily operations for Mel-Airport
  - 500+ aircraft land and depart
  - Every one needs a parking lot
  - Grounds crew to turn it around
- You are the operation planner:
  - **Which gate** each flight parks at
  - **When** each ground task runs
  - **Which team** performs which task

Hackathon: Optimisation Challenge

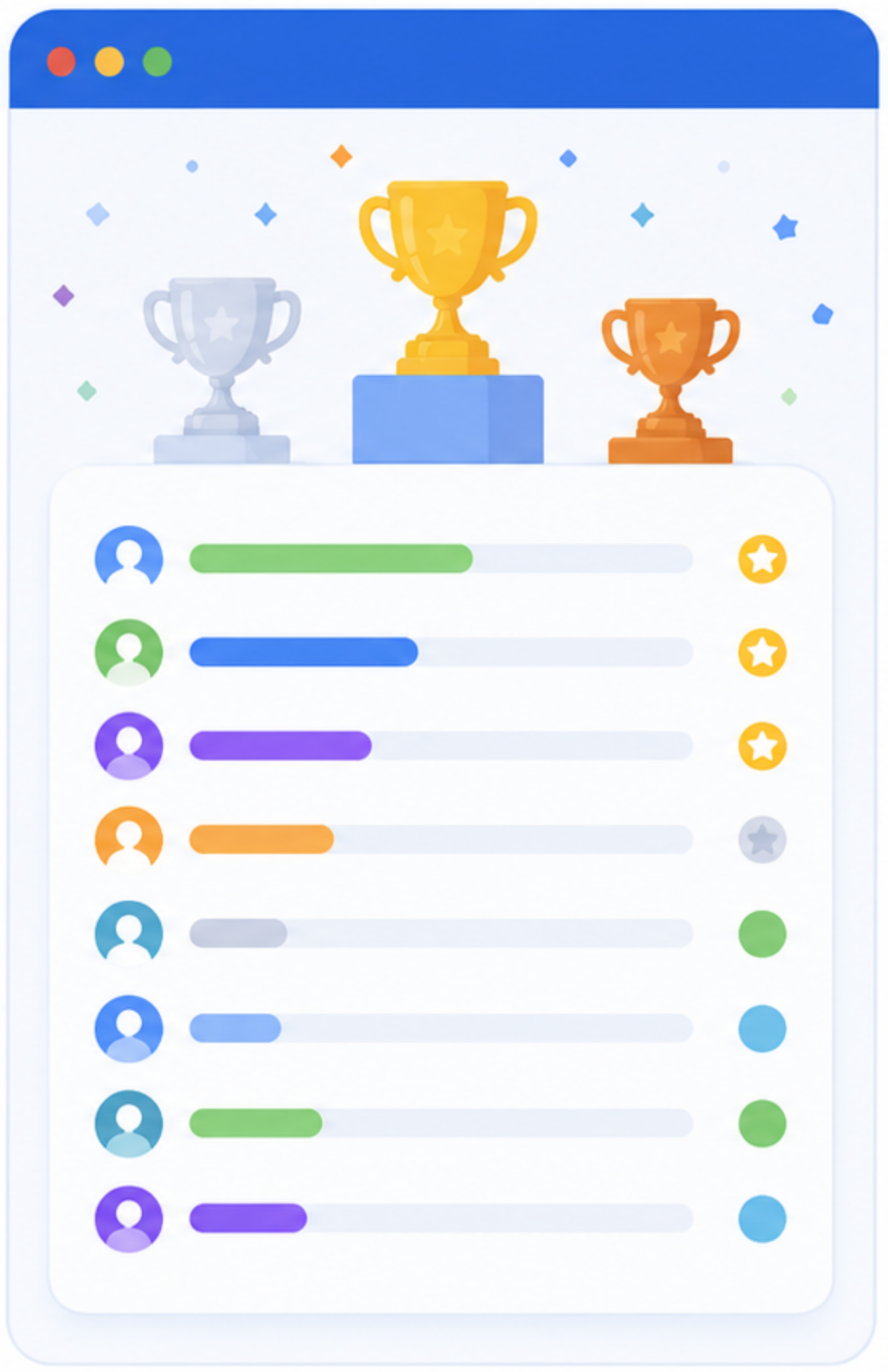
- Website: https://hackathon.minizinc.dev/



Submission



Debug



Leaderboard

Hackathon: Submission

ACP Summer School 2026 Hackathon Submission Portal

Submit | Leaderboard | Debug

Submissions close in **3d 04:43:14**

Deadline **28 Aug 2026, 16:00**

Leaderboard freeze **28 Aug 2026, 15:30**

Get the bundle

Everything you need to start solving the problem.

- hackathon.pdf — problem specification
- data — folder with instance files
- Project.mzp — a MiniZinc project file
- airport_sched.mzn — starter MiniZinc model
- starter.py — starter Python script

Download bundle (.zip)

Submit solutions

Drop one or many `sol_XX.json` files. We infer the instance from the filename and grade each one independently. No team token yet? Sign up here, we'll email you a token.

Team token

team-...

Emailed to your team after signing up via the form.

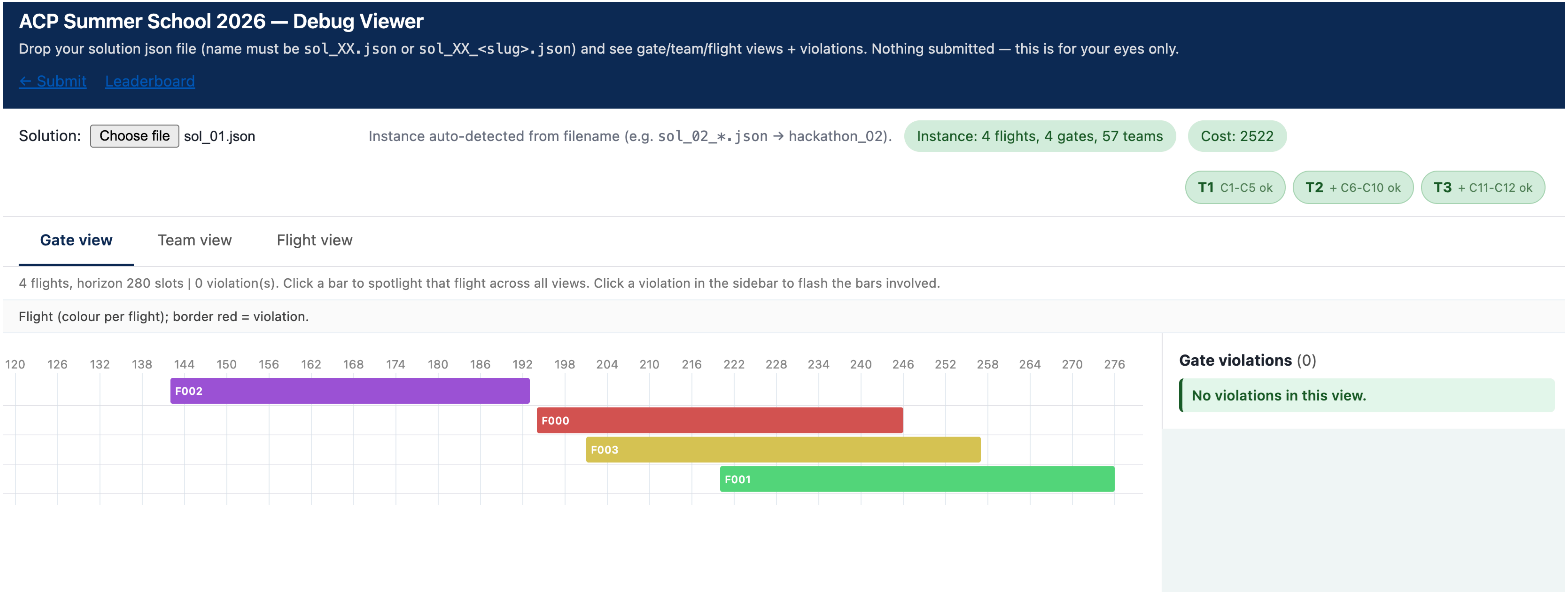
Solution files

Drop `sol_XX.json` files here, or click to browse

Multiple files OK. Instance ID auto-detected from filename (e.g. `sol_06.json` → hackathon_06)

Grade 0 file(s)

Hackathon: Debug



Hackathon: Leaderboard

Submissions close in **15:43:17** | Deadline **29 Jul 2026, 15:30** | Freeze **29 Jul 2026, 15:00**

Standings — points per instance (higher is better, out of 100), total out of 1000

Updated 23:46:36 · 4 teams

#	TEAM	01	02	03	04	05	06	07	08	09	10	TOTAL
1 🥇	**The Flying Kangaroo** @monash	100.0 T3	20.0 T1	—	100.0 T3	—	91.6 T3	100.0 T3	100.0 T3	100.0 T3	—	**611.6**
2 🥈	**Melbourne Run!** @melbourne	—	100.0 T3	—	89.9 T3	—	97.7 T3	96.9 T3	97.5 T3	—	—	**482.0**
3 🥉	**A Good Day** @rmit	—	20.0 T1	100.0 T3	—	100.0 T3	97.7 T3	—	97.5 T3	—	—	**415.2**
4	**Crazyflight** @mel-uni	—	20.0 T1	100.0 T3	89.9 T3	—	97.7 T3	—	—	—	—	**307.6**

Highest points on this instance · Tier 3 (correct) · Tier 2 (violates C10/C11) · Tier 1 (gate only) · — = no submission

Hackathon: Instance Points

- The points for each of 12 instances is calculated individually:

	Constraints	Problem Scope	Points formula
Tier 1	C1 to C5	Gate assignment	20
Tier 2	C1 to C10	+ Tasks assignment and scheduling	+ 50 x (best cost / your cost)
Tier 3	C1 to C12	+ Travel and rest constraints	+ 30 x (best cost / your cost)

- Instance points = Tier 1 points + Tier 2 points + Tier 3 points

Let’s Get Start!

- Find your teammates and sign up!
- Download the bundle in the website: https://hackathon.minizinc.dev/
- Work on gate assignment solutions!

Good Luck!
The apron is yours!

Today: Gate Assignment

- Flights have attributes:
  - Arrival time and departure times
  - Wide/narrow-body, passenger/freighter, international/domestic flights
- Gates have attributes:
  - Open time and close times
  - Large/small stand, passenger/freighter, international/domestic gates
- Handle uncertainty: 30-minute between flights at the same gate
- Hint: think about the “alldifferent” global constraint

Today: Data Format

Listing 1: Instance JSON

```
{
  "horizon":    1440,
  "FlightID":   ["JST917", "UAE9314", "JST034", "MXD177", "CPA3128"],
  "GateID":     ["D02", "D03", "D04", "H03"],
  "LaborID":    ["ramp_team_01", "ramp_team_02", "baggage_team_01",
                 /* ... one per team unit ... */],
  "max_tasks": 11,

  /* Data for flights */
  "flight_arr": [31, 754, 242, 256, 855],
  "flight_dep": [76, 889, 287, 301, 990],
  "flight_carrier": ["domestic","international","domestic",
                     "domestic","domestic"],
  "flight_size": ["narrow","wide","narrow","narrow","wide"],
  "flight_op_type": ["passenger","passenger","passenger",
                     "passenger","freighter"],
```

The number of minutes

Today: Data Format

Listing 1: Instance JSON

Array indexed by FlightID

```
{
  "horizon":   1440,
  "FlightID":  ["JST917", "UAE9314", "JST034", "MXD177", "CPA3128"],
  "GateID":    ["D02", "D03", "D04", "H03"],
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  "flight_op_type": ["passenger","passenger","passenger",
                     "passenger","freighter"],
```

Flight JST917 lands at 00:31
and departs at 01:16

Today: Data Format

Listing 1: Instance JSON

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{
  "horizon":   1440,
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  "GateID":    ["D02", "D03", "D04", "H03"],
  "LaborID":   ["ramp_team_01", "ramp_team_02", "baggage_team_01",
                /* ... one per team unit ... */],
```

Array indexed by GateID

```
/* Data for gates */
"gate_stand_size": ["small","large","large","large"],
"gate_usage":      ["domestic","international","international","domestic"],
"gate_op_type":    ["passenger","passenger", "passenger","freighter"],
"gate_open":       [0, 0, 0, 360],
"gate_close":      [1440, 1440, 1440, 1440],
```

Tier 2 & 3: Task scheduling

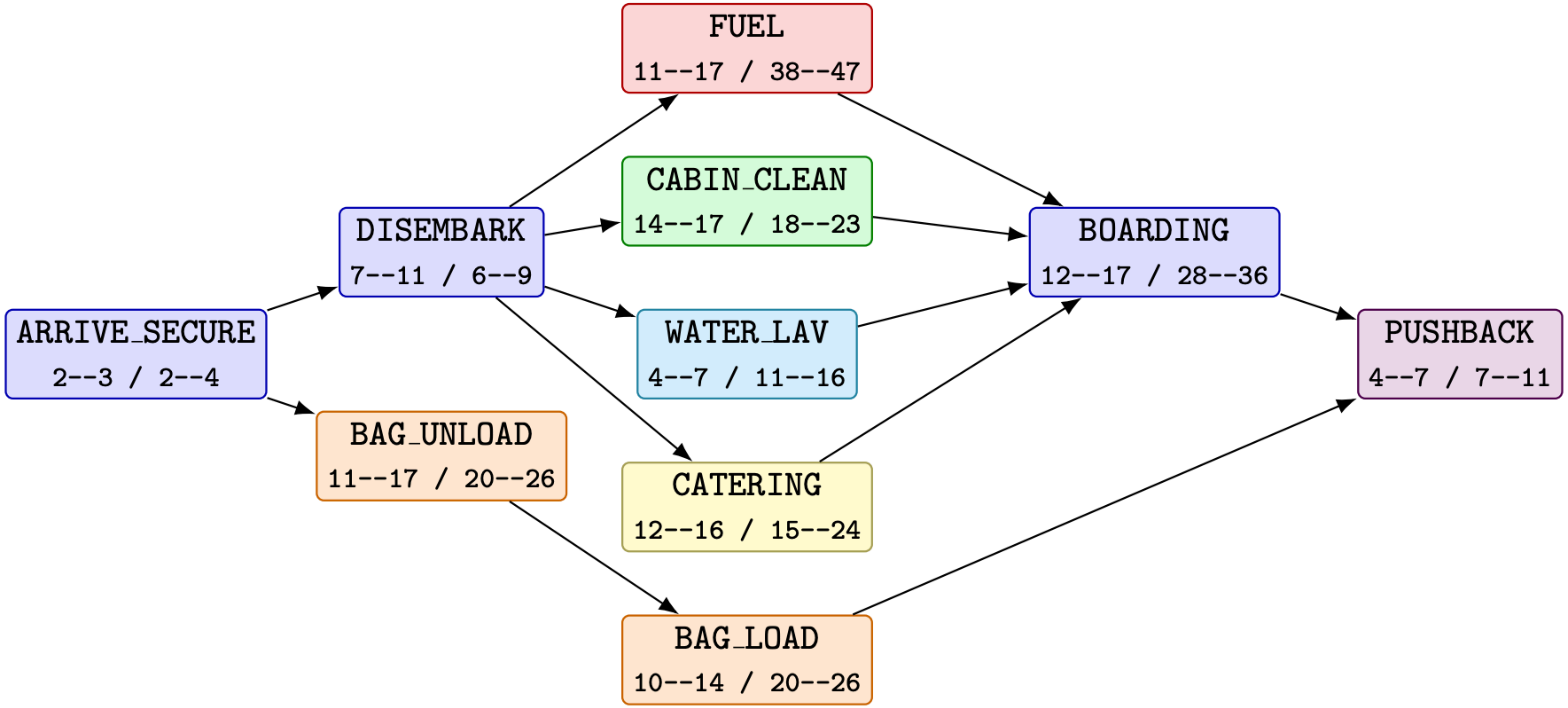

Figure 1: **Domestic passenger profile**. Durations in minutes.

Tier 2 & 3: Task scheduling

"flight_n_tasks": [10, 11, 10, 10, 4],
"flight_tasks": [
[{"kind":"ARRIVE_SECURE","duration":2}, {"kind":"DISEMBARK","duration":7},
{“kind”:”BAG_UNLOAD","duration":13}, {“kind”:”FUEL","duration": 15},
{“kind”:”CARBIN_CLEAN”,"duration":16}, {“kind”:”WATER_LAV”,”duration":5},
{“kind”:”CATERING”,"duration":15}, {“kind”:”BAG_LOAD”,”duration”:12},
{“kind”:”BOARDING”,"duration":15}, {“kind”:”PUSHBACK”,”duration”:6},
{“kind”:”PUSHBACK”,"duration":0}],
/* ... one padded row per flight ... */],

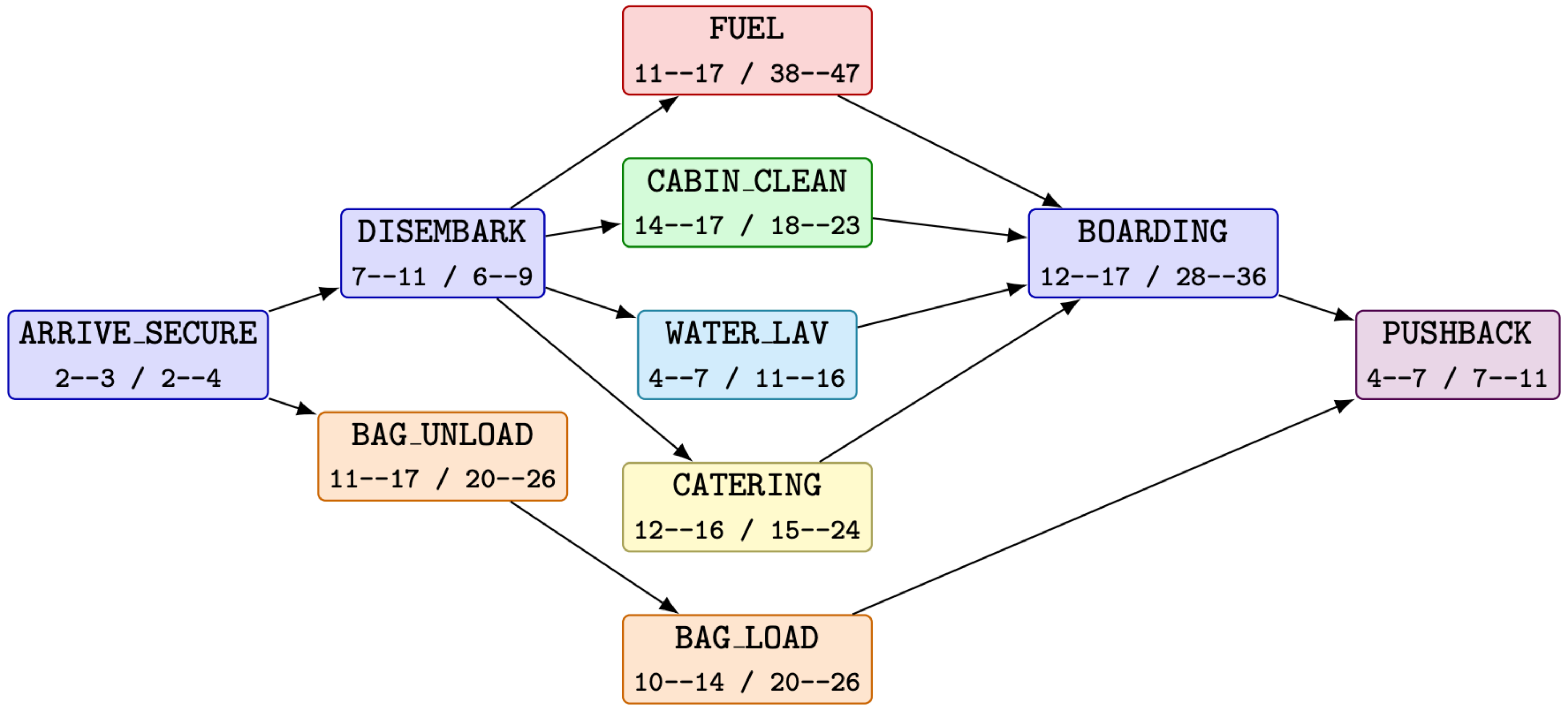

Figure 1: **Domestic passenger profile**. Durations in minutes.

Tier 2 & 3: Task scheduling

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{“kind”:”CARBIN_CLEAN”,"duration":16}, {“kind”:”WATER_LAV”,”duration":5},
{“kind”:”CATERING”,"duration":15}, {“kind”:”BAG_LOAD”,”duration”:12},
{“kind”:”BOARDING”,"duration":15}, {“kind”:”PUSHBACK”,”duration”:6}, …],
/* ... one padded row per flight ... */],

"task_start": [[31, 33, 33, 40, 40, 40, 40, 44, 54, 66, 0],
/* ... one row per flight, padded to max_tasks ... */],
"task_labor": [["ramp_team_01", "ramp_team_01", "baggage_team_01", "fuel_team_01",
"carbin_clean_team_01", "water_waste_team_01", "catering_team_01",
“baggage_team_01", "ramp_team_01", "pushback_team_01", "ramp_team_01"],
/* ... one row per flight ... */],
"cost": 4231

Tier 2 & 3: Task scheduling

- Visualise your solution: https://hackathon.minizinc.dev/debug.html
- Some techniques that may be useful:
  - global constraints: all_different, disjunctive, cumulative
  - symmetry breaking & dominance breaking
  - large neighbourhood search
  - separate problem into different stages
  - and more?

A bit update on leaderboard

ACP Summer School 2026 Hackathon Live Leaderboard

Submit | Leaderboard | Debug

Submissions close in **1d 22:09:54** | Deadline **28 Aug 2026, 15:30** | Freeze **28 Aug 2026, 14:00**

Standings — cost per instance (lower is better)

Updated 17:19:59 · 12 teams

		01–07 · SMALL							08–12 · LARGE					
#	TEAM	01	02	03	04	05	06	07	08	09	10	11	12	TOTAL
1	LZL ACP	2,063 T2	4,491 T2	6,799 T2	8,312 T2	10,930 T2	14,382 T2	11,879 T2	17,933 T2	21,134 T2	22,003 T2	74,752 T2	66,193 T2	840.0
2	做不出来去睡觉 睡饱了	2,522 T3	4,491 T3	8,333 T3	8,312 T2	10,930 T2	14,382 T2	11,879 T2	17,933 T2	22,925 T2	24,421 T2	— T1	— T1	802.8
3	Avalon Airpot Team JNU-JLU-XJTU	2,063 T2	4,491 T2	6,799 T2	8,312 T2	12,163 T2	19,549 T2	12,132 T2	22,861 T2	24,455 T2	25,372 T2	118,504 T2	116,982 T2	756.3
4	Ox is here Not here yet	2,522 T3	4,491 T3	8,975 T3	11,367 T3	15,283 T3	23,755 T3	17,157 T3	37,680 T3	37,292 T3	—	—	—	751.8

Sharing Sessions

- Time: Friday 15:30 to 16:30
- Rule for micro talk:
  - At most 5 minutes per team
  - At most 3 slides per talk (+ title card)
- Explain the idea of your team’s solution:
  - What solver/tool did you use?
  - How long did you run for each instance?
  - What techniques did you use and which one you find useful?
- Send me your slides before the deadline